

Question-4.6.10

EE25BTECH11022 - sankeerthan

Question

Find the equation of the line passing through the point $(1, -3, 2)$ and parallel to the line $\mathbf{r} = (2 + \lambda)\hat{i} + \lambda\hat{j} + (-1 + \lambda)\hat{k}$

Solution

Given line is

$$\mathbf{r} = \begin{pmatrix} 2 + \lambda \\ \lambda \\ -1 + \lambda \end{pmatrix} \quad \mathbf{a} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} \quad (1)$$

The vector equation of given line is given by

$$\mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (2)$$

The direction vectors of given line are

$$\mathbf{m} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (3)$$

Solution

The lines with direction vectors $\mathbf{m}$ and $\mathbf{n}$ are parallel if

$$\mathbf{m} = \mathbf{n} \implies \mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (4)$$

The equation of a line is given by

$$\mathbf{r} = \mathbf{a} + t\mathbf{n} \quad (5)$$

$$\mathbf{r} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (6)$$

3D Line passing through $(1, -3, 2)$

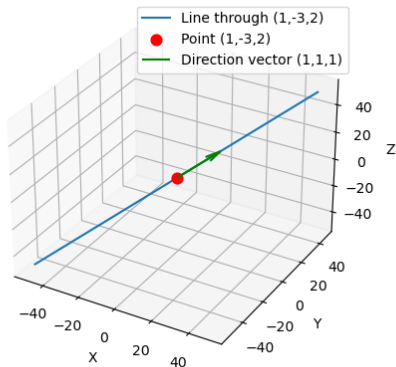


Figure:

C-Code

```
// code.c
#include <stdio.h>

void line_points(double* xs, double* ys, double* zs, int n) {
    double x0 = 1, y0 = -3, z0 = 2; // given point
    double dx = 1, dy = 1, dz = 1; // direction vector
    for (int i = 0; i < n; ++i) {
        double lambda = (i - n/2); // range  $[-n/2, n/2]$ 
        xs[i] = x0 + lambda * dx;
        ys[i] = y0 + lambda * dy;
        zs[i] = z0 + lambda * dz;
    }
}

void normal_vector(double* normal) {
    normal[0] = 1;
    normal[1] = 1;
    normal[2] = 1;
}
```

```
# Code by GVV Sharma  
# Modified for Problem Solution  
# Released under GNU GPL  
# Calculating area enclosed between curves  
# call.py  
# call.py  
import ctypes  
import numpy as np  
  
N = 100  
xs = np.zeros(N)  
ys = np.zeros(N)  
zs = np.zeros(N)  
normal = np.zeros(3)
```

```
1 lib = ctypes.cdll.LoadLibrary('./line.so')
2 arrtype = np.ctypeslib.ndpointer(dtype=np.float64, ndim=1, flags=
3     'C_CONTIGUOUS')
4
5 lib.line_points.argtypes = [arrtype, arrtype, arrtype, ctypes.
6     c_int]
7 lib.line_points(xs, ys, zs, N)
8
9 lib.normal_vector.argtypes = [arrtype]
10 lib.normal_vector(normal)
11
12 np.savez('points.npz', xs=xs, ys=ys, zs=zs, normal=normal)
```



```
# plot.py
import numpy as np
import matplotlib.pyplot as plt

data = np.load('points.npz')
xs, ys, zs = data['xs'], data['ys'], data['zs']
normal = data['normal']

fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

# Line
ax.plot(xs, ys, zs, label="Line through (1, -3, 2)")
```

```
# Point
ax.scatter([1], [-3], [2], color='red', s=60, label="Point
            (1,-3,2)")

# Normal vector (as direction vector)
ax.quiver(1, -3, 2, normal[0], normal[1], normal[2], color='green
            ', length=15, label="Direction vector (1,1,1)")

ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_zlabel('Z')
ax.legend()
plt.title('3D Line and its Direction Vector')
plt.show()
```